Source: Sahil Bhola, Self-contained Proof for a Mixed-precision Block Jacobi Algorithm.

Selected deliverable: the full active result labeled theorem:block_jacobi_real in main.tex (lines 613-693).

The original mathematical proof is available in paper.tex. The selected statement below is authoritative.
Earlier statements and notation supply context. A referenced result may be proved as a helper. Do not assume a
desired algorithmic bound or an unproved paper lemma as an added hypothesis.

SELECTED STATEMENT (verbatim LaTeX, proof omitted)

```
\begin{theorem}[Update-form Block Jacobi in Real Arithmetic]
\label{theorem:block_jacobi_real}
Let $\mat{a}\ireal{n\times n}$ be decomposed as
$\mat{a}=\mat{d}+\mat{n}$, where $\mat{d}$ is a nonsingular block-diagonal
matrix and $\mat{n}$ contains the off-diagonal blocks. Let
$\vect{x}$ satisfy $\mat{a}\vect{x}=\vect{b}$, and consider the exact
block-Jacobi iteration
\begin{align}
    \mat{d}\vect{c}^{k+1}
    &=\vect{b}-\mat{a}\vect{x}^k,
    &
    \vect{x}^{k+1}
    &=\vect{x}^k+\vect{c}^{k+1}.
\end{align}
Define the forward error $\vect{e}^k\define\vect{x}^k-\vect{x}$ and the
iteration matrix $\mat{p}\define-\mat{d}^{-1}\mat{n}$. Then
\begin{align}
    \vect{e}^{k+1}=\mat{p}\vect{e}^k,
    \qquad
    \vect{e}^k=\mat{p}^k\vect{e}^0,
\end{align}
and hence
\begin{align}
    \abs{\vect{e}^k}
    &\leq \abs{\mat{p}}^k\abs{\vect{e}^0},
    &
    \norm[\infty]{\vect{e}^k}
    &\leq
    \norm[\infty]{\mat{p}}^k
    \norm[\infty]{\vect{e}^0}.
\end{align}
The iteration converges to $\vect{x}$ for every initial guess if and only
if
\begin{align}
    \rho(\mat{p})
    =\rho(\mat{d}^{-1}\mat{n})<1.
\end{align}
In particular, $\norm[\infty]{\mat{p}}<1$ is a sufficient condition for
convergence and makes the iteration a contraction in the infinity norm.
\end{theorem}
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SECTION CONTEXT

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\section{Block Jacobi Error Analysis in Reals}
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SOURCE NOTATION MACROS

```
\newcommand{\urd}{\ensuremath{\mathrm{u}}}
\newcommand{\define}{\ensuremath{\triangleq}}
\newcommand{\vect}[1]{\mathbf{\MakeLowercase{#1}}}
\newcommand{\mat}[1]{\mathbf{\MakeUppercase{#1}}}
\newcommand{\ireal}[1]{\in\mathbb{R}^{#1}}
\newcommand{\norm}[2][]{\left\|#2\right\|_{#1}}
\DeclareMathOperator{\fl}{fl}
\newcommand{\R}{\mathbb{R}}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\ceil}[1]{\left\lceil #1 \right\rceil}
\newcommand{\widehatgamma}[1]{\widehat{\gamma}_{#1}}
```